

Final Model Performance Report – Credit Card Fraud Detection

1. Developed Model Overview

Objective:

The model was created to find credit card transactions by sorting transactions into:

Class 0: Transactions that are not fraud

Class 1: Transactions that are fraud

The final model was made better by adjusting parameters and tested on data that was not used before.

2. Final Model Performance Summary

Metric	Performance
Accuracy	~99%
Fraud Detection Recall	~91%
Legitimate Transaction Detection Recall	~98.5%
Fraud Precision	~10%
Fraud F1-score	~17%

3. Areas Where the Model Is Performing Well

3.1 Strong Fraud Detection Capability

The model found about 91% of transactions.

This means the model is good at finding activity and not missing many fraud cases.

The small number of fraud cases that were marked as shows that the model is working well.

Business Impact:

Saves money by stopping fraud before it causes damage.

Helps teams look at the dangerous transactions.

3.2 High Accuracy in Identifying Legitimate Transactions

The model got 98.5% of the real transactions right.

Fewer false alarms for transactions.

Business Impact:

Keeps customers from being upset.

Saves time by not checking transactions.

3.3 Stable Overall Classification Performance

The model had 98.5% accuracy overall.

It is good at telling the difference between fake transactions.

4. Areas Where the Model Requires Improvement

4.1 Low Fraud Prediction Precision

Current Performance:

Fraud precision: 10%

Issue:

Many transactions that the model says are fraud are actually real.

This creates alerts.

Business Impact:

work for people who check for fraud.

4.2 Improving Balance Between Recall and Precision

Current Situation:

The model tries to find all the fraud, which gives recall.

4.3 Further Model Enhancement

Possible improvements:

Need to try better machine learning models:

Random Forest

XGBoost

LightGBM

Ensemble models

Work on better features to find:

How transactions happen

How customers spend money

When transactions happen over time

5. Overall Model Assessment

The final Logistic Regression model for fraud detection is very good at finding fraud with recall and high accuracy. It does a job at finding most fraud and not messing up real transactions.

The main area to improve is making mistakes when saying a transaction is fraud. This will make the system better for banks.

KEY FINDINGS

The developed Logistic Regression model was effective in detecting most fraudulent transactions, achieving high recall and accuracy. However, improving fraud precision and reducing false positives are the main areas for further enhancement before real-world implementation.

Project Summary

This project aimed to develop a credit card fraud detection model using a dataset of credit card transactions. The process involved key stages

Data Loading and Initial Inspection

The dataset was successfully loaded from Google Drive using gdown. Then read into a pandas DataFrame. Initial inspection using `credit_df.head()` `credit_df.info()` and `credit_df.describe()` revealed that the dataset contained 284,807 entries with 31 columns all of which were numerical. Crucially it was confirmed that there were no missing values across any of the columns

Exploratory Data Analysis (EDA) and Data Cleaning

Histograms and Box Plots Visualizations of feature distributions revealed that most V features were already scaled likely due to PCA transformation. However Time and Amount had distributions and outliers

Feature Scaling

To handle the outliers in Time and Amount RobustScaler was applied to these two features making them more robust for modeling

Class Imbalance

A significant class imbalance was identified with only 0.17% of transactions being fraudulent. This necessitated balancing techniques for model training

Feature Correlation

Correlation analysis with the Class variable showed varying degrees of relationships. V17 had the negative correlation while V11 had the strongest positive correlation

Feature Engineering

A new feature Hour was extracted from the Time column showing a slightly improved correlation with the Class compared to the raw Time feature

Model Development and Evaluation

The dataset was split into training and testing sets (80/20 split) using `train_test_split` with `stratify=y` to maintain the class distribution

To address the severe class imbalance RandomUnderSampler was applied to the training data resulting in a balanced training set with an equal number of fraudulent and non-fraudulent transactions (394 each)

Model Selection

Logistic Regression was chosen as the classification algorithm

Model Training

A baseline Logistic Regression model was trained on the data

Hyperparameter Tuning

GridSearchCV was employed to find the optimal hyperparameters for the Logistic Regression model. The best parameters found were $C=0.1$ $\text{max_iter}=1000$ $\text{penalty}='l1'$ and $\text{solver}='liblinear'$ yielding a score of approximately 0.948 on the undersampled data

Model Evaluation

The best model was then evaluated on the original imbalanced test set (with a probability threshold of 0.7). The key evaluation metrics were

Accuracy 0.98517

Precision 0.09622

Recall 0.90816

F1-Score 0.17400

ROC AUC Score 0.97827

Confusion Matrix Analysis

The model demonstrated recall (detecting 90.8% of actual fraud cases) and high specificity (correctly approving 98.5% of normal transactions). However it also exhibited precision (flagging, about 1.5% of genuine transactions as fraud) indicating a significant number of false positives. This highlights a trade-off in fraud detection prioritizing the detection of fraud (recall) often comes at the cost of increased false alarms (lower precision). Further work could focus on refining this balance